FedAda: Towards Efficient Client-Edge-Cloud Federated Learning with Adaptive Communication - Proof of Theorem 1

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Assumption 1 (Lipschitz Gradient)**.** *There exists a constant $L > 0$, such that:*

$$\|\nabla F(\boldsymbol{\omega}_1) - \nabla F(\boldsymbol{\omega}_2)\| \leq L\|\boldsymbol{\omega}_1 - \boldsymbol{\omega}_2\|,\ \forall \boldsymbol{\omega}_1, \boldsymbol{\omega}_2 \tag{1}$$

Assumption 2 (Unbiased Estimator)**.** *Let ξ be a random sample from dataset and the estimated gradient is unbiased:*

$$\mathbb{E}\left[\nabla F(\boldsymbol{\omega}; \xi)\right] = \nabla F(\boldsymbol{\omega}) \tag{2}$$

Assumption 3 (Bounded Variance)**.** *There exists a constant σ, such that the variance of the gradient can be bounded by:*

$$\mathbb{E}\left[\|\nabla F(\boldsymbol{\omega}; \xi) - \nabla F(\boldsymbol{\omega})\|^2\right] \leq \sigma^2 \tag{3}$$

Before stating the proof of Theorem 1 of FedAda, we first mention the following three lemmas.

Lemma 1. *According to Assumption 1, the expected inner product between full batch gradient and stochastic gradient can be bounded with:*

$$\begin{aligned}
&\mathbb{E}\left\langle \nabla f\left(\boldsymbol{\omega}^h\right), \frac{1}{N}\sum_{i=1}^{M}\alpha_i^h \sum_{r=1}^{k_2}\sum_{j\in\mathcal{V}_i}\alpha_j^h \sum_{t=1}^{k_1}\nabla F_j\left(\boldsymbol{\omega}_j^{t,r,h}\right)\right\rangle \\
&\leq -\frac{1}{2N}\sum_{i=1}^{M}\alpha_i^h \sum_{r=1}^{k_2}\sum_{j\in\mathcal{V}_i}\alpha_j^h \sum_{t=1}^{k_1}\left(\|\nabla f\left(\boldsymbol{\omega}^h\right)\|^2 + \|\nabla F_j\left(\boldsymbol{\omega}_j^{t,r,h}\right)\|^2\right) \\
&+ \frac{L^2}{2N}\sum_{i=1}^{M}\alpha_i^h \sum_{r=1}^{k_2}\sum_{j\in\mathcal{V}_i}\alpha_j^h \sum_{t=1}^{k_1}\|\boldsymbol{\omega}^h - \boldsymbol{\omega}_j^{t,r,h}\|^2
\end{aligned} \tag{4}$$

Proof.

$$\begin{aligned}
&\mathbb{E}\left\langle \nabla f\left(\boldsymbol{\omega}^h\right), \frac{1}{N}\sum_{i=1}^{M}\alpha_i^h \sum_{r=1}^{k_2}\sum_{j\in\mathcal{V}_i}\alpha_j^h \sum_{t=1}^{k_1}\nabla F_j\left(\boldsymbol{\omega}_j^{t,r,h}\right)\right\rangle \\
&= \frac{1}{N}\sum_{i=1}^{M}\alpha_i^h \sum_{r=1}^{k_2}\sum_{j\in\mathcal{V}_i}\alpha_j^h \sum_{t=1}^{k_1}\left\langle \nabla f\left(\boldsymbol{\omega}^h\right), \nabla F_j\left(\boldsymbol{\omega}_j^{t,r,h}\right)\right\rangle \\
&\overset{①}{=} \frac{1}{2N}\sum_{i=1}^{M}\alpha_i^h \sum_{r=1}^{k_2}\sum_{j\in\mathcal{V}_i}\alpha_j^h \sum_{t=1}^{k_1}\left[-\|\nabla f\left(\boldsymbol{\omega}^h\right)\|^2 - \|\nabla F_j\left(\boldsymbol{\omega}_j^{t,r,h}\right)\|^2 + \|\nabla f\left(\boldsymbol{\omega}^h\right) - \nabla F_j\left(\boldsymbol{\omega}_j^{t,r,h}\right)\|^2\right]
\end{aligned} \tag{5}$$

$$\overset{②}{\leq} \frac{1}{2N}\sum_{i=1}^{M}\alpha_i^h\sum_{r=1}^{k_2}\sum_{j\in\mathcal{V}_i}\alpha_j^h\sum_{t=1}^{k_1}\left[-\|\nabla f\left(\boldsymbol{\omega}^h\right)\|^2-\|\nabla F_j\left(\boldsymbol{\omega}_j^{t,r,h}\right)\|^2+L^2\|\boldsymbol{\omega}^h-\boldsymbol{\omega}_j^{t,r,h}\|^2\right]$$

where ① is due to $2\langle\boldsymbol{x},\boldsymbol{y}\rangle=\|\boldsymbol{x}\|^2+\|\boldsymbol{y}\|^2-\|\boldsymbol{x}-\boldsymbol{y}\|^2$, and ② based on Assumption.1. □

Lemma 2. *Under Assumption 2 and Assumption 3, we have the following bound:*

$$\mathbb{E}\|g^h\|^2\leq\frac{1}{N^2}\sum_{i=1}^{M}\left(\alpha_i^h\right)^2\sum_{r=1}^{k_2}\sum_{j\in\mathcal{V}_i}\left(\alpha_j^h\right)^2\sum_{t=1}^{k_1}\left[k_1\|\nabla F_j\left(\boldsymbol{\omega}_j^{t,r,h}\right)\|^2+\sigma^2\right] \tag{6}$$

Proof. Let $\sum_{t=1}^{k_1}\nabla F_j\left(\boldsymbol{\omega}_j^{t,r,h}\right)$ be $\tilde{F}_j$. Then we have:

$$\begin{aligned}
&\mathbb{E}\|\frac{1}{N}\sum_{i=1}^{M}\alpha_i^h\sum_{r=1}^{k_2}\sum_{j\in\mathcal{V}_i}\alpha_j^h\sum_{t=1}^{k_1}\nabla F_j\left(\boldsymbol{\omega}_j^{t,r,h}\right)\|^2=\mathbb{E}\|\frac{1}{N}\sum_{i=1}^{M}\alpha_i^h\sum_{r=1}^{k_2}\sum_{j\in\mathcal{V}_i}\alpha_j^h\nabla\tilde{F}_j\|^2\\
&\overset{①}{=}Var\left(\frac{1}{N}\sum_{i=1}^{M}\alpha_i^h\sum_{r=1}^{k_2}\sum_{j\in\mathcal{V}_i}\alpha_j^h\nabla\tilde{F}_j\right)+\|\frac{1}{N}\sum_{i=1}^{M}\alpha_i^h\sum_{r=1}^{k_2}\sum_{j\in\mathcal{V}_i}\alpha_j^h\nabla\tilde{F}_j\|^2\\
&\leq\frac{1}{N^2}\left(\alpha_i^h\alpha_j^h\right)^2\sum_{i=1}^{M}\sum_{r=1}^{k_2}\sum_{j\in\mathcal{V}_i}Var\left(\nabla\tilde{F}_j\right)+\frac{1}{N^2}\left(\alpha_i^h\alpha_j^h\right)^2\sum_{i=1}^{M}\sum_{r=1}^{k_2}\sum_{j\in\mathcal{V}_i}\nabla\tilde{F}_j\\
&\overset{②}{\leq}\frac{1}{N^2}\left(\alpha_i^h\alpha_j^h\right)^2\sum_{i=1}^{M}\sum_{r=1}^{k_2}\sum_{j\in\mathcal{V}_i}k_1\sigma^2+\frac{1}{N^2}\left(\alpha_i^h\alpha_j^h\right)^2\sum_{i=1}^{M}\sum_{r=1}^{k_2}\sum_{j\in\mathcal{V}_i}\nabla\tilde{F}_j\\
&\overset{③}{\leq}\frac{1}{N^2}\sum_{i=1}^{M}\left(\alpha_i^h\right)^2\sum_{r=1}^{k_2}\sum_{j\in\mathcal{V}_i}\left(\alpha_j^h\right)^2\sum_{t=1}^{k_1}\left[k_1\|\nabla F_j\left(\boldsymbol{\omega}_j^{t,r,h}\right)\|^2+\sigma^2\right]
\end{aligned} \tag{7}$$

where ① is due to $\mathbb{E}\left[\|x\|^2\right]=Var\left(x\right)+\|\mathbb{E}\left[x\right]\|^2$ and follows Assumption.2, ② follows Assumption.3 and has $Var\left(\nabla\tilde{F}_j\right)=Var\left(\sum_{t=1}^{k_1}\nabla F_j\left(\boldsymbol{\omega}_j^{t,r,h}\right)\right)=\sum_{t=1}^{k_1}Var\left(\nabla F_j\left(\boldsymbol{\omega}_j^{t,r,h}\right)\right)\leq k_1\sigma^2$, and ③ is because $\|\nabla\tilde{F}_j\|^2=\|\sum_{t=1}^{k_1}\nabla F_j\left(\boldsymbol{\omega}_j^{t,r,h}\right)\|^2\leq k_1\sum_{t=1}^{k_1}\|\nabla F_j\left(\boldsymbol{\omega}_j^{t,r,h}\right)\|^2$. □

Lemma 3. *Under Assumptions 3 and according to [1], we have the following bound:*

$$\mathbb{E}\|\boldsymbol{\omega}^h-\boldsymbol{\omega}_j^{t,r,h}\|^2\leq\eta^2\left[\left(\frac{k_2-1}{Nk_2}\right)^2+\frac{(k_1-1)^2}{Nk_1^2k_2^2}\right]\sum_{i=1}^{M}\sum_{r=1}^{k_2}\sum_{j\in\mathcal{V}_i}\sum_{t=1}^{k_1}\left[k_1\|\nabla F_j\left(\boldsymbol{\omega}_j^{t,r,h}\right)\|^2+\sigma^2\right] \tag{8}$$

Proof. According to [1], we have the following relationship:

$$\frac{1}{N}\sum_{j=1}^{N}\|\nabla F_j\left(\boldsymbol{\omega}\right)-\nabla f\left(\boldsymbol{\omega}\right)\|^2=\sum_{i=1}^{M}\frac{N_i}{N}\|\nabla f_i\left(\boldsymbol{\omega}\right)-\nabla f\left(\boldsymbol{\omega}\right)\|^2+\sum_{i=1}^{M}\frac{1}{N}\sum_{j\in\mathcal{V}_i}\|\nabla F_j\left(\boldsymbol{\omega}\right)-\nabla f_i\left(\boldsymbol{\omega}\right)\|^2 \tag{9}$$

Under the Law of Large Numbers, we have $\mathbb{E}\left[\boldsymbol{\omega}^h-\eta\nabla f\left(\boldsymbol{\omega}^h\right)\right]=\mathbb{E}\left[\boldsymbol{\omega}_j^{t,r,h}-\eta\nabla f\left(\boldsymbol{\omega}_j^{t,r,h}\right)\right]$, so that we have $\|\boldsymbol{\omega}^h-\boldsymbol{\omega}_j^{t,r,h}\|^2=\eta^2\|\nabla F_j\left(\boldsymbol{\omega}\right)-\nabla f\left(\boldsymbol{\omega}\right)\|^2$, and then we can get:

$$\frac{1}{N}\sum_{j=1}^{N}\|\boldsymbol{\omega}^h-\boldsymbol{\omega}_j^{t,r,h}\|^2=\eta^2\sum_{i=1}^{M}\frac{N_i}{N}\|\nabla f_i\left(\boldsymbol{\omega}\right)-\nabla f\left(\boldsymbol{\omega}\right)\|^2+\eta^2\sum_{i=1}^{M}\frac{1}{N}\sum_{j\in\mathcal{V}_i}\|\nabla F_j\left(\boldsymbol{\omega}\right)-\nabla f_i\left(\boldsymbol{\omega}\right)\|^2 \tag{10}$$

Thus, we can prove the Lemma.3 by

$$
\begin{aligned}
(1)\quad & \sum_{i=1}^{M}\frac{N_i}{N}\|\nabla f_i\left(\boldsymbol{\omega}\right)-\nabla f\left(\boldsymbol{\omega}\right)\|^2 \\
&=\sum_{i=1}^{M}\frac{N_i}{N}\|\frac{1}{N_i}\sum_{j\in\mathcal{V}_i}\sum_{t=1}^{k_1}\nabla F_j\left(\boldsymbol{\omega}_j^{t,r,h}\right)-\frac{1}{N}\sum_{i=1}^{M}\sum_{r=1}^{k_2}\sum_{j\in\mathcal{V}_i}\sum_{t=1}^{k_1}\nabla F_j\left(\boldsymbol{\omega}_j^{t,r,h}\right)\|^2 \\
&=\sum_{i=1}^{M}\frac{N_i}{N}\|\frac{1}{Nk_2}\sum_{i=1}^{M}\sum_{r=1}^{k_2}\sum_{j\in\mathcal{V}_i}\sum_{t=1}^{k_1}\nabla F_j\left(\boldsymbol{\omega}_j^{t,r,h}\right)-\frac{1}{N}\sum_{i=1}^{M}\sum_{r=1}^{k_2}\sum_{j\in\mathcal{V}_i}\sum_{t=1}^{k_1}\nabla F_j\left(\boldsymbol{\omega}_j^{t,r,h}\right)\|^2 \\
&=\sum_{i=1}^{M}\frac{N_i}{N}\|\frac{k_2-1}{Nk_2}\sum_{i=1}^{M}\sum_{r=1}^{k_2}\sum_{j\in\mathcal{V}_i}\sum_{t=1}^{k_1}\nabla F_j\left(\boldsymbol{\omega}_j^{t,r,h}\right)\|^2 \\
&\overset{①}{\leq}\sum_{i=1}^{M}\frac{N_i}{N}\left\{\left(\frac{k_2-1}{Nk_2}\right)^2\sum_{i=1}^{M}\sum_{r=1}^{k_2}\sum_{j\in\mathcal{V}_i}\left[Var\left(\nabla\tilde{F}_j\right)+\|\nabla\tilde{F}_j\|^2\right]\right\} \\
&\overset{②}{\leq}\left(\frac{k_2-1}{Nk_2}\right)^2\sum_{i=1}^{M}\sum_{r=1}^{k_2}\sum_{j\in\mathcal{V}_i}\sum_{t=1}^{k_1}\left[k_1\|\nabla F_j\left(\boldsymbol{\omega}_j^{t,r,h}\right)\|^2+\sigma^2\right]
\end{aligned}
\tag{11}
$$

$$
\begin{aligned}
(2)\quad & \sum_{i=1}^{M}\frac{1}{N}\sum_{j\in\mathcal{V}_i}\|\nabla F_j\left(\boldsymbol{\omega}\right)-\nabla f_i\left(\boldsymbol{\omega}\right)\|^2 \\
&=\sum_{i=1}^{M}\frac{1}{N}\sum_{j\in\mathcal{V}_i}\|\nabla F_j\left(\boldsymbol{\omega}_j^{t,r,h}\right)-\frac{1}{N_i}\sum_{j\in\mathcal{V}_i}\sum_{t=1}^{k_1}\nabla F_j\left(\boldsymbol{\omega}_j^{t,r,h}\right)\|^2 \\
&=\sum_{i=1}^{M}\frac{1}{N}\sum_{j\in\mathcal{V}_i}\|\frac{1}{Nk_1k_2}\sum_{i=1}^{M}\sum_{r=1}^{k_2}\sum_{j\in\mathcal{V}_i}\sum_{t=1}^{k_1}\nabla F_j\left(\boldsymbol{\omega}_j^{t,r,h}\right)-\frac{1}{Nk_2}\sum_{i=1}^{M}\sum_{r=1}^{k_2}\sum_{j\in\mathcal{V}_i}\sum_{t=1}^{k_1}\nabla F_j\left(\boldsymbol{\omega}_j^{t,r,h}\right)\|^2 \\
&=\sum_{i=1}^{M}\frac{1}{N}\sum_{j\in\mathcal{V}_i}\|\frac{k_1-1}{Nk_1k_2}\sum_{i=1}^{M}\sum_{r=1}^{k_2}\sum_{j\in\mathcal{V}_i}\sum_{t=1}^{k_1}\nabla F_j\left(\boldsymbol{\omega}_j^{t,r,h}\right)\|^2 \\
&\overset{①}{\leq}\sum_{i=1}^{M}\frac{1}{N}\sum_{j\in\mathcal{V}_i}\left(\frac{k_1-1}{Nk_1k_2}\right)^2\sum_{i=1}^{M}\sum_{r=1}^{k_2}\sum_{j\in\mathcal{V}_i}\left[Var\left(\nabla\tilde{F}_j\right)+\|\nabla\tilde{F}_j\|^2\right] \\
&\overset{②}{\leq}N\left(\frac{k_1-1}{Nk_1k_2}\right)^2\sum_{i=1}^{M}\sum_{r=1}^{k_2}\sum_{j\in\mathcal{V}_i}\sum_{t=1}^{k_1}\left[k_1\|\nabla F_j\left(\boldsymbol{\omega}_j^{t,r,h}\right)\|^2+\sigma^2\right]
\end{aligned}
$$

Combining (1) and (2), we can obtain:

$$
\begin{aligned}
&\frac{1}{N}\sum_{j=1}^{N}\|\boldsymbol{\omega}^h-\boldsymbol{\omega}_j^{t,r,h}\|^2=\eta^2\sum_{i=1}^{M}\frac{N_i}{N}\|\nabla f_i\left(\boldsymbol{\omega}\right)-\nabla f\left(\boldsymbol{\omega}\right)\|^2+\eta^2\sum_{i=1}^{M}\frac{1}{N}\sum_{j\in\mathcal{V}_i}\|\nabla F_j\left(\boldsymbol{\omega}\right)-\nabla f_i\left(\boldsymbol{\omega}\right)\|^2 \\
&\leq\eta^2\left[\left(\frac{k_2-1}{Nk_2}\right)^2+\frac{(k_1-1)^2}{Nk_1^2k_2^2}\right]\sum_{i=1}^{M}\sum_{r=1}^{k_2}\sum_{j\in\mathcal{V}_i}\sum_{t=1}^{k_1}\left[k_1\|\nabla F_j\left(\boldsymbol{\omega}_j^{t,r,h}\right)\|^2+\sigma^2\right]
\end{aligned}
\tag{12}
$$

where ① and ② refer to the Lemma.2. □

Now, based on the above assumptions and lemmas, let's prove Theorem 1. First, according to Assump-

tion.1 and the SGD model update formula we can obtain:

$$
\begin{aligned}
\mathbb{E}\left\{f\left(\boldsymbol{\omega}^{h+1}\right)\right\} &= \mathbb{E}\left\{f\left(\boldsymbol{\omega}^{h}-\eta\frac{1}{N}\sum_{i=1}^{M}\alpha_i^h\sum_{r=1}^{k_{2,i}^h}\sum_{j\in\mathcal{V}_i}\alpha_j^h\sum_{t=1}^{k_{1,j}^h}\nabla F_j\left(\boldsymbol{\omega}_j^{t,r,h}\right)\right)\right\} \\
&\leq \mathbb{E}\left\{f\left(\boldsymbol{\omega}^{h}-\eta\frac{1}{N}\sum_{i=1}^{M}\alpha_1\sum_{r=1}^{k_2}\sum_{j\in\mathcal{V}_i}\alpha_2\sum_{t=1}^{k_1}\nabla F_j\left(\boldsymbol{\omega}_j^{t,r,h}\right)\right)\right\} \\
&\leq \mathbb{E}\left\{f\left(\boldsymbol{\omega}^{h}\right)\right\}-\eta\mathbb{E}\left\{\left\langle\nabla f\left(\boldsymbol{\omega}^{h}\right),\frac{1}{N}\sum_{i=1}^{M}\alpha_1\sum_{r=1}^{k_2}\sum_{j\in\mathcal{V}_i}\alpha_2\sum_{t=1}^{k_1}\nabla F_j\left(\boldsymbol{\omega}_j^{t,r,h}\right)\right\rangle\right\} \\
&+\frac{\eta^2L}{2}\mathbb{E}\|\frac{1}{N}\sum_{i=1}^{M}\alpha_1\sum_{r=1}^{k_2}\sum_{j\in\mathcal{V}_i}\alpha_2\sum_{t=1}^{k_1}\nabla F_j\left(\boldsymbol{\omega}_j^{t,r,h}\right)\|^2
\end{aligned}
\tag{13}
$$

where $k_1=\max\left\{k_{1,j}^h\right\}$, $k_2=\max\left\{k_{2,i}^h\right\}$, $\alpha_1=\max\left\{\alpha_j^h\right\}$, $\alpha_2=\max\left\{\alpha_i^h\right\}$.

Then, based on the Lemma.1, Lemma.2, Lemma.3 and (13), we can obtain:

$$
\begin{aligned}
&\mathbb{E}\left\{f\left(\boldsymbol{\omega}^{h+1}\right)-f\left(\boldsymbol{\omega}^{h}\right)\right\} \\
&\leq\frac{\eta}{2N}\sum_{i=1}^{M}\alpha_1\sum_{r=1}^{k_2}\sum_{j\in\mathcal{V}_i}\alpha_2\sum_{t=1}^{k_1}\left[-\|\nabla f\left(\boldsymbol{\omega}^{h}\right)\|^2-\|\nabla F_j\left(\boldsymbol{\omega}_j^{t,r,h}\right)\|^2+L^2\|\boldsymbol{\omega}^h-\boldsymbol{\omega}_j^{t,r,h}\|^2\right] \\
&+\frac{\eta^2L}{2}\frac{1}{N^2}\sum_{i=1}^{M}\alpha_1^2\sum_{r=1}^{k_2}\sum_{j\in\mathcal{V}_i}\alpha_2^2\sum_{t=1}^{k_1}\left[k_1\|\nabla F_j\left(\boldsymbol{\omega}_j^{t,r,h}\right)\|^2+\sigma^2\right] \\
&\leq\frac{\eta}{2N}\sum_{i=1}^{M}\alpha_1\sum_{r=1}^{k_2}\sum_{j\in\mathcal{V}_i}\alpha_2\sum_{t=1}^{k_1}\left[-\|\nabla f\left(\boldsymbol{\omega}^{h}\right)\|^2-\|\nabla F_j\left(\boldsymbol{\omega}_j^{t,r,h}\right)\|^2\right] \\
&+\frac{L^2\eta^3}{2N}\frac{\left[k_1^2\left(k_2-1\right)^2+N\left(k_1-1\right)^2\right]}{Nk_1k_2}\alpha_1\alpha_2\sum_{i=1}^{M}\sum_{r=1}^{k_2}\sum_{j\in\mathcal{V}_i}\sum_{t=1}^{k_1}\left[k_1\|\nabla F_j\left(\boldsymbol{\omega}_j^{t,r,h}\right)\|^2+\sigma^2\right] \\
&+\frac{L\eta^2}{2N^2}\alpha_1^2\alpha_2^2\sum_{i=1}^{M}\sum_{r=1}^{k_2}\sum_{j\in\mathcal{V}_i}\sum_{t=1}^{k_1}\left[k_1\|\nabla F_j\left(\boldsymbol{\omega}_j^{t,r,h}\right)\|^2+\sigma^2\right] \\
&=-\frac{\eta\alpha_1\alpha_2k_1k_2}{2}\|\nabla f\left(\boldsymbol{\omega}^{h}\right)\|^2+\frac{L^2\eta^3\alpha_1\alpha_2\left[k_1^2\left(k_2-1\right)^2+N\left(k_1-1\right)^2\right]}{2N}\sigma^2+\frac{L\eta^2\alpha_1^2\alpha_2^2k_1k_2}{2N}\sigma^2 \\
&-\left\{\frac{\eta\alpha_1\alpha_2}{2N}-\frac{L^2\eta^3\alpha_1\alpha_2\left[k_1^2\left(k_2-1\right)^2+N\left(k_1-1\right)^2\right]}{2N^2k_2}-\frac{L\eta^2\alpha_1^2\alpha_2^2k_1}{2N^2}\right\}\sum_{i=1}^{M}\sum_{r=1}^{k_2}\sum_{j\in\mathcal{V}_i}\sum_{t=1}^{k_1}\|\nabla F_j\left(\boldsymbol{\omega}_j^{t,r,h}\right)\|^2 \\
&=-\frac{\eta\alpha_1\alpha_2k_1k_2}{2}\|\nabla f\left(\boldsymbol{\omega}^{h}\right)\|^2+\frac{L\eta^2\alpha_1\alpha_2}{2N}\left\{L\eta\left[k_1^2\left(k_2-1\right)^2+N\left(k_1-1\right)^2\right]+\alpha_1\alpha_2k_1k_2\right\}\sigma^2 \\
&-\frac{\eta\alpha_1\alpha_2}{2N}\left\{1-\frac{L^2\eta^2\left[k_1^2\left(k_2-1\right)^2+N\left(k_1-1\right)^2\right]}{Nk_2}-\frac{L\eta\alpha_1\alpha_2k_1}{N}\right\}\sum_{i=1}^{M}\sum_{r=1}^{k_2}\sum_{j\in\mathcal{V}_i}\sum_{t=1}^{k_1}\|\nabla F_j\left(\boldsymbol{\omega}_j^{t,r,h}\right)\|^2 \\
&\leq-\frac{\eta\alpha_1\alpha_2k_1k_2}{2}\|\nabla f\left(\boldsymbol{\omega}^{h}\right)\|^2+\frac{L\eta^2\alpha_1\alpha_2}{2N}\left\{L\eta\left[k_1^2\left(k_2-1\right)^2+N\left(k_1-1\right)^2\right]+\alpha_1\alpha_2k_1k_2\right\}\sigma^2
\end{aligned}
\tag{14}
$$

where k_1, k_2, α_1, α_2 and η satisfy:

$$\frac{L^2\eta^2\left[k_1^2\left(k_2-1\right)^2+N\left(k_1-1\right)^2\right]}{Nk_2}-\frac{L\eta\alpha_1\alpha_2k_1}{N}\leq 1 \tag{15}$$

Next, based on (14), we can obtain:

$$\begin{aligned}
&\frac{1}{H}\sum_{h=1}^{H}\|\triangledown f\left(\boldsymbol{\omega}^h\right)\|^2\\
&\leq\frac{2\left[f\left(\boldsymbol{\omega}^0\right)-f\left(\boldsymbol{\omega}^*\right)\right]}{\eta\alpha_1\alpha_2k_1k_2H}+\frac{L\eta}{Nk_1k_2}\left\{L\eta\left[k_1^2\left(k_2-1\right)^2+N\left(k_1-1\right)^2\right]+\alpha_1\alpha_2k_1k_2\right\}\sigma^2\\
&\leq\frac{2\left[f\left(\boldsymbol{\omega}^0\right)-f\left(\boldsymbol{\omega}^*\right)\right]}{\eta\alpha_1\alpha_2k_1k_2H}+\frac{L\eta}{Nk_1k_2}\left\{L\eta\left[k_1^2k_2^2+Nk_1^2\right]+\alpha_1\alpha_2k_1k_2\right\}\sigma^2\\
&\leq\frac{2\left[f\left(\boldsymbol{\omega}^0\right)-f\left(\boldsymbol{\omega}^*\right)\right]}{\eta\alpha_1\alpha_2k_1k_2H}+\frac{L\eta\alpha_1\alpha_2}{N}\sigma^2+\frac{L^2\eta^2k_1k_2}{N}\sigma^2+\frac{L^2\eta^2k_1}{k_2}\sigma^2
\end{aligned} \tag{16}$$

Thus, the proof is completed.

References

[1] Jiayi Wang, Shiqiang Wang, Rong-Rong Chen, et al. “Demystifying why local aggregation helps: Convergence analysis of hierarchical SGD”. In: *AAAI*. Vol. 36. 8. 2022, pp. 8548–8556.